

Fysik 1 10060

Forårs-forelæsning #13 *Energitema 1*

Med Noter



Forelæser:
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Adjunk, DTU Fysik

Kursusoverblik

Overblik over forårets forelæsninger, **foreløbig** plan:

Uge 1: Kinematik 1D (Kap. 3)

Uge 2: Kinematik 2D (Kap. 4)

Uge 3: Usikkerhed

Uge 4: Eksperimenter

Uge 5: Kræfter 1 (Kap. 5)

Uge 6: Kræfter 2 (Kap. 6)

Uge 7: Bevægelsesligninger (Kap. 15)

Uge 8: Arbejde Og Energi (Kap. 7)

Påske (30/3)

Påske (6/4)

Uge 9: Energibevarelse (Kap. 8)

Uge 10: Stød Og Impulsbevarelse (Kap. 9)

Uge 11: Rotation 1 (Kap. 10)

Uge 12: Rotation 2 (Kap. 11)

Uge 13: Energitema 1

Dagens forelæsning:

Uge 13: Energitema



- The scales of world energy
- Energy Storage
- Flywheel (svinghjul)
 - Kinetic energy
 - Stress/strain
 - Limitations

Repetition (Kinematik 1D): Relevante formler

Konstant acceleration

Equation	Contains
$v = v_0 + at$	v, a, t ; no x
$x = x_0 + \frac{1}{2}(v_0 + v)t$	x, v, t ; no a
$x = x_0 + v_0t + \frac{1}{2}at^2$	x, a, t ; no v
$v^2 = v_0^2 + 2a(x - x_0)$	x, v, a ; no t

Frit fald acceleration

$$v = v_0 - gt \text{ (positive upward)}$$

$$y = y_0 + v_0t - \frac{1}{2}gt^2 \text{ (positive upward)}$$

General acceleration

$$\frac{dv}{dt} = a \Rightarrow v(t) = \int_0^t a(t)dt + v_0$$

$$\frac{dx}{dt} = v \Rightarrow x(t) = \int_0^t v(t)dt + x_0$$

Repetition (Kinematik 2D): Relevante formler

- Projektilbevægelse (skrå kast)

Affyring/
Landing
Horizontal

Tid $T_{tof} = \frac{2v_0 \sin(\theta_0)}{g}$

Bane ligning: $y = \tan(\theta_0) x - \left[\frac{g}{2v_0^2 \cos^2(\theta)} \right] x^2$

Højeste punkt: $y = \frac{(v_0 \sin(\theta))^2}{2g}$

Længde: $R = \frac{v_0^2 \sin(2\theta_0)}{g}$

Generelt

$x(t) = v_{0x} t$
 $y(t) = v_{0y} t - \frac{1}{2} g t^2$
 $v_x(t) = v_{0x}$
 $v_y(t) = v_{0y} - g t$

$v_y^2 = v_{0y}^2 - 2g(y - y_0)$
 $v^2 = v_0^2 - 2g(y - y_0)$
 $y(x) = y_0 + \tan(\theta) x - \frac{g}{2v_0^2 \cos^2(\theta)} x^2$



- Cirkel bevægelse

$$v = \frac{2\pi r}{T}$$

$$a_c = \frac{v^2}{r}$$

$$a_T = \frac{d|v|}{dt}$$

- Relativ bevægelse

$$v_{PS} = v_{PS'} + v_{S'S}$$

$$a_{PS} = a_{PS'} + a_{S'S}$$

- Trigonometriske formler

Repetition (Usikkerhed): Relevante formler

Angivelse af usikkerhed

x	bedste bud på størrelse
δx	usikkerhed, positiv størrelse
$x \pm \delta x$	absolut usikkerhed
$\frac{\delta x}{ x }$	relativ usikkerhed

Måleserie

Standardafvigelsen er $\delta x = \sqrt{\frac{\sum(\bar{x} - x_i)^2}{N-1}}$ (målingerne er en stikprøve)

Usikkerheden på gennemsnittet, $\bar{x}$, er $\delta \bar{x} = \frac{\delta x}{\sqrt{N}}$

-> standard deviation of the mean (SDOM).

Normal fordeling

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$$P(\mu - \delta x \leq x \leq \mu + \delta x) = 0.68$$

$$P(\mu - 2\delta x \leq x \leq \mu + 2\delta x) = 0.95$$

$$P(\mu - 3\delta x \leq x \leq \mu + 3\delta x) = 0.99$$

Tilfældige og systematiske usikkerheder

$$\delta x = \sqrt{(\delta x_{\text{tilfældige}})^2 + (\delta x_{\text{systematiske}})^2}$$

Beregninger med størrelser med usikkerhed

Beregn $z = x \pm y$

Uafhængige usikkerheder: $z \pm \delta z = x \pm y \pm \sqrt{\delta x^2 + \delta y^2}$

Afhængige usikkerheder: $z \pm \delta z = x \pm y \pm \delta x + \delta y$

Beregn $z = x \cdot y$

Uafhængige usikkerheder (relativ) $\frac{\delta z}{|z|} = \sqrt{\left(\frac{\delta x}{|x|}\right)^2 + \left(\frac{\delta y}{|y|}\right)^2}$

Afhængige usikkerheder (relativ) $\frac{\delta z}{|z|} = \frac{\delta x}{|x|} + \frac{\delta y}{|y|}$

Fejlophobningsloven

$z = f(x, y), x \pm \delta x, y \pm \delta y$

Uafhængige usikkerheder: $\delta z = \sqrt{\left(\frac{\partial f}{\partial x} \delta x\right)^2 + \left(\frac{\partial f}{\partial y} \delta y\right)^2}$

Afhængige usikkerheder: $\delta z = \left|\frac{\partial f}{\partial x}\right| \delta x + \left|\frac{\partial f}{\partial y}\right| \delta y$

Repetition (Eksperimenter): Relevante formler

$$\text{stdscore} = \frac{x - \bar{x}}{\delta x}$$

$\text{stdscore} < 2$ *Good*

$2 < \text{stdscore} < 3$ *Grey zone*

$3 < \text{stdscore}$ *reject*

Sammenligning af to målinger

$$\text{stdscore} = \frac{|z|}{\delta z}$$

$$z = x - y$$

$$\delta z = \sqrt{\delta x^2 + \delta y^2}$$

Naturlige skalaer

$\mu = m$ (masse)

$\lambda = l$ (længde)

$\tau = \sqrt{l/g}$ (tidsdimensionen)

$\alpha = \lambda/\tau^2 = l/(l/g)$ (acceleration)

$t = f(m, g, h, k)$

Model

	m	g	h	k
M	1	0	0	1
L	0	1	1	0
T	0	-2	0	-1

Dimensionsmatricen

$v_2 = f(v_1, a, x)$

Model

$\frac{v_2}{v_1} = F\left(\frac{ax}{v_1^2}\right)$

Dimensionsløs formulering af modellen

Repetition (Newtons love): Relevante formler

Newtons love:

N1:

$$\sum F = 0 \Rightarrow v = \textit{konstant}$$

N2:

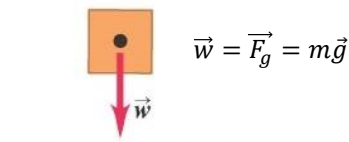
$$\sum F = ma$$

N3:

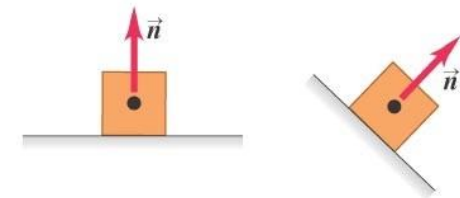
$$\overrightarrow{F_{AB}} = -\overrightarrow{F_{BA}}$$

Fem vigtige kræfter i mekanik

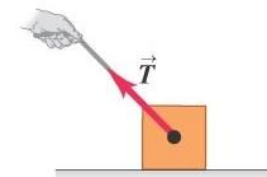
Tyngdekraft



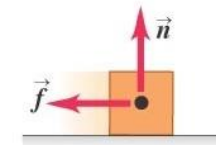
Normalkraft



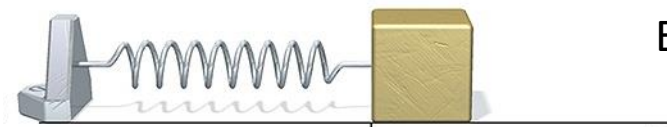
Snorkraft



Gnidningskraft



Fjederkraft



Enheden for kraft: $N = \text{kg} \cdot \text{m/s}^2$

Repetition (kræfter 2): Relevante formler

- **Hooke's lov for fjederkraften:**

$$F_{\text{fjeder}} = -k \cdot (x - x_0)$$

$$f_k = \mu_k n \quad \text{Kinematisk gnidning}$$

$$f_s \leq \mu_s n \quad \text{Statisk gnidning}$$

$$F_d = b \cdot v^n \quad \text{Luft- og vandmodstand (generelt)}$$

$$f = D \cdot v^2 \quad \text{Luft- og vandmodstand (drag)}$$

$$D = \frac{1}{2} \cdot \rho_{\text{luft}} \cdot C_D \cdot A$$

- **Rundt object der falder i et medium (drag)**

$$F_s = 6\pi r \eta v \quad \text{Stokes' law}$$

- **Cirkel bevægelse**

$$\vec{r} = (x, y) = (r \cos \theta, r \sin \theta)$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

$$\omega = \frac{d\theta}{dt}$$

$$v = r \omega$$

$$\frac{d\omega}{dt} = \alpha$$

$$a_t = \frac{dv}{dt} = r \alpha \quad \text{Bemærk: } a_t = 0 \text{ ved cirkelbevægelse med konstant fart!}$$

$$a_c = v \omega = r \omega^2 = \frac{v^2}{r}$$

$$v = \frac{\text{omkreds}}{\text{periode}} = \frac{2\pi R}{T}$$

Repetition (bevægelsesligninger): Relevante formler

- Eulers metode

$$x(t + \Delta t) \approx x(t) + v(t)\Delta t$$

$$v(t + \Delta t) \approx v(t) + a(t)\Delta t$$

- Eulers-Cromers metode

$$v(t + \Delta t) \approx v(t) + a(t)\Delta t$$

$$x(t + \Delta t) \approx x(t) + v(t + \Delta t)\Delta t$$

- Scipy solve_ivp

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.integrate import solve_ivp
4 t1 = 0.0
5 t2 = 10.0
6 m = 1.0
7 k = 10.0
8 def f(t,y):
9     x,v = y
10    return [v, -k*x/m]
11 sol = solve_ivp(f,[t1,t2],y0=[0.1,0.0],atol=1e-8,rtol=1e-8)
12 t = sol.t
13 x = sol.y[0]
14 v = sol.y[1]
```

- Events

```
def hit_ground(t,z):
    y,v = z
    return y
```

```
hit_ground.terminal = True
hit_ground.direction = -1
```

Repitition (Energibevarelse): Relevante formler

Arbejde

$$W = \vec{F} \cdot \vec{s}$$

Kinetisk Energi

$$E = \frac{1}{2}mv^2$$

Potentiel Energi

$$\Delta E = mg\Delta h$$

Fjederkraft (Hookes lov):

$$F_{\text{fjeder}} = -kx$$

Arbejde fra en fjederkraft:

$$\begin{aligned} W_{\text{fjeder}} &= \int_{x_1}^{x_2} F_{\text{fjeder}}(x) dx \\ &= \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 \end{aligned}$$

Effekt

$$P = \frac{W}{\Delta t} = F \frac{\Delta s}{\Delta t} = F v$$

Repetition (Stød og Impulsbevarelse): Relevante formler

Impuls

$$\vec{p} = m\vec{v}$$

Kinetisk energi på impulsform

$$K = \frac{p^2}{2m}$$

Newtons 2. lov udtrykt ved impuls

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

(1) Impulsbevarelse

$$m_A v_{A1} + m_B v_{B1} = m_A v_{A2} + m_B v_{B2}$$

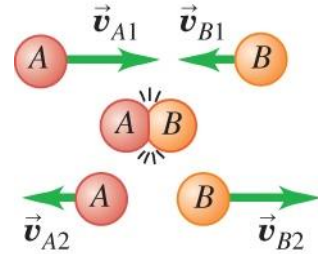
(2) Energibevarelsen

$$v_{B2} - v_{A2} = -(v_{B1} - v_{A1})$$

3 forskellige typer af kollisioner:

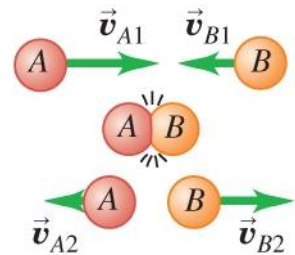
Elastisk kollision

- Impulsbevarelse
- Bevarelse af kinetisk energi



Uelastisk kollision

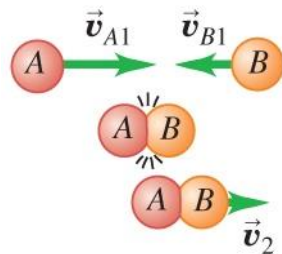
- Impulsbevarelse



Fuldstændig uelastisk kollision

- Impulsbevarelse
- De to legemer har fælles hastighed før eller efter stødet.

Elastisk



Repetition (Rotation 1): Relevante formler

Rotationsenergi

$$K_{rot} = \frac{1}{2} I \omega^2$$

Kinetisk energi af bevægende og roterende legeme

$$K = K_{rot} + K_{trans} = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} M v_{cm}^2$$

cm = center of mass

Energi bevarelse

$$K_{1,trans} + K_{1,rot} + U_1 + W_{other} = K_{2,trans} + K_{2,rot} + U_2$$

Parallelaksætningen

$$I_{||} = I_{cm} + M d^2$$

Inertimomenter

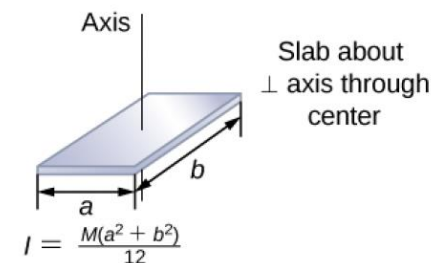
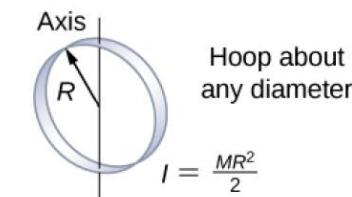
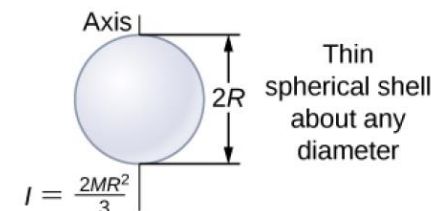
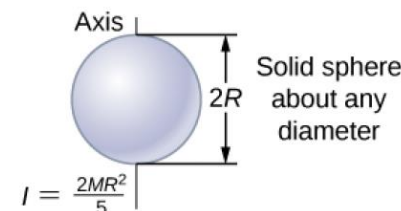
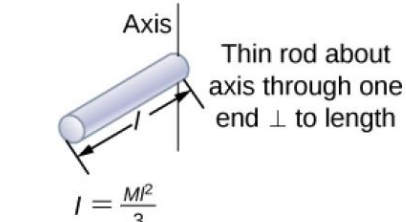
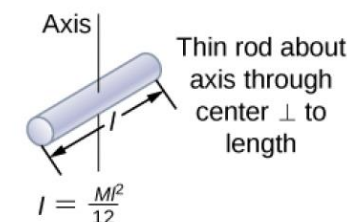
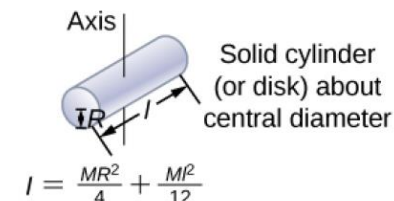
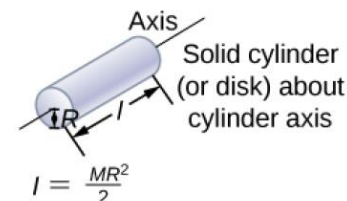
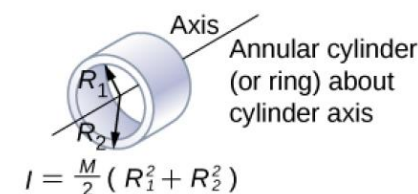
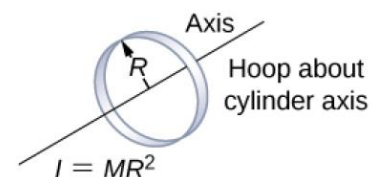


Figure 10.20 Values of rotational inertia for common shapes of objects.

Repetition (Rotation 1): Relevante formler

Kraftmoment

$$\rightarrow \vec{\tau} = \vec{R} \times \vec{F}$$

Newtons anden lov for rotation
(impulsmomentsætningen IMS)

$$\rightarrow I_{\text{cm}} \alpha = \sum \tau_{\text{cm}} \quad M \vec{a}_{\text{cm}} = \sum \vec{F}_{\text{ydre}}$$

Geometrisk bånd
(Ren rulning):

$$\underline{x_{\text{cm}} = R\theta}, \quad \underline{v_{\text{cm}} = R\omega}, \quad \underline{a_{\text{cm}} = R\alpha}$$

Generelt

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

Impulsmoment (vinkel moment)

$$\left[\begin{array}{l} \vec{L} = \vec{R} \times \vec{p} = R_{\perp} p \\ L = I\omega \end{array} \right.$$

Bevarelse af Impulsmoment

$$\sum \tau = 0$$

$$\Rightarrow \frac{dL}{dt} = 0$$

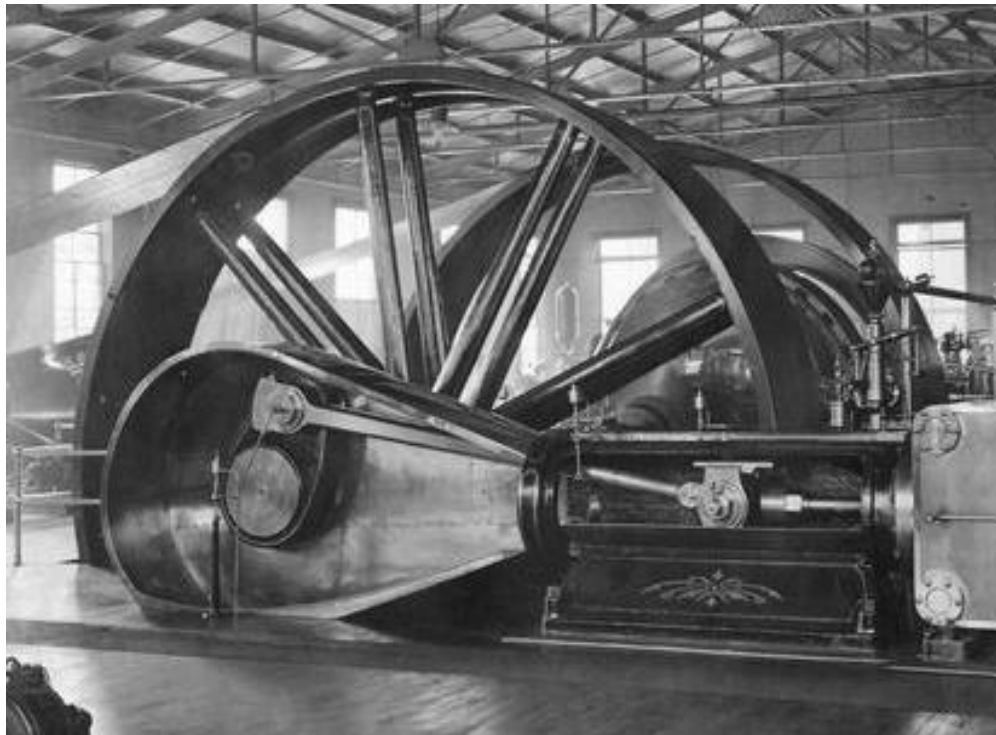
$$\Rightarrow L = l_1 + l_2 + \dots = \textit{konstant}$$

Eksempel:

$$\hookrightarrow I_1 \omega_1 = I_2 \omega_2$$

Physical energy storage: Flywheels

Ca 1890: Steel flywheel for steam engines



2022: Grid frequency stabilization via 3 MW flywheel (ABB/ S4 power, Netherlands)





Rank the energy sources with the largest first

Coal

Hydropower

Gas

Oil

Renewables (wind/solar)

Nuclear





Rank the energy sources with the largest first

Correct order:

1. Oil
2. Coal
3. Gas
4. Nuclear
5. Hydropower
6. Renewables (wind/solar)

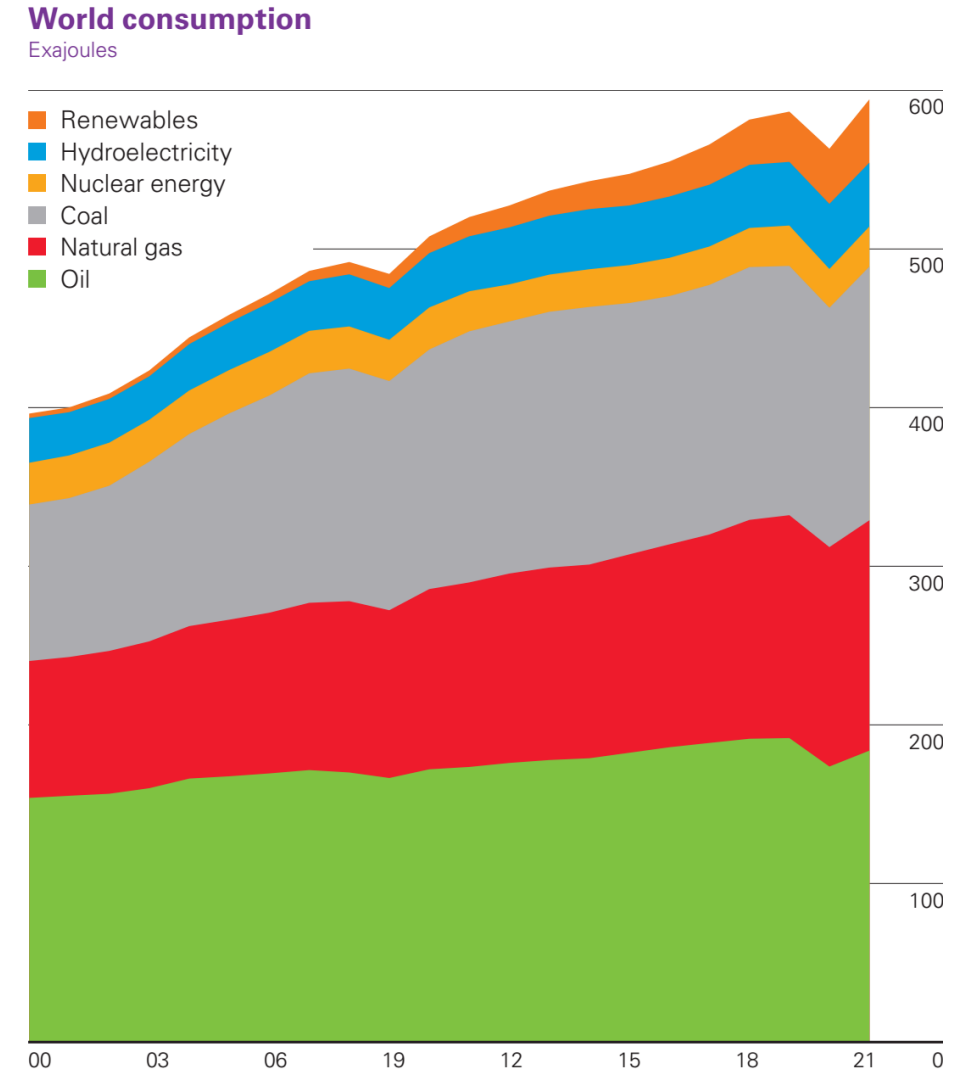
Correct responses: 3%

RESULTS SLIDE

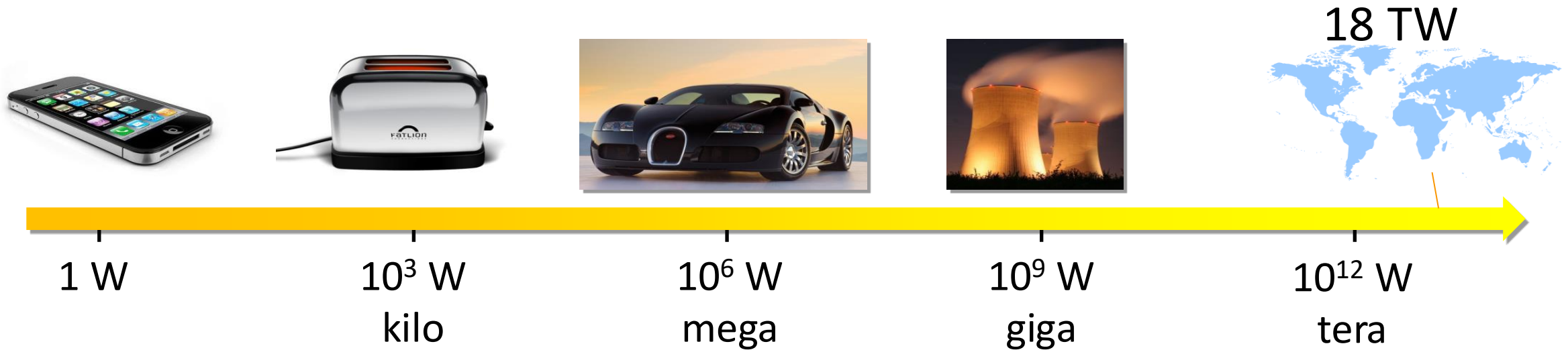
Setting the stage: World energy

- Total energy: 570 EJ/yr = 18 TW
 - Oil: 31.2%
 - Coal: 27.2%
 - Gas: 24.7%
 - Nuclear: 4.3%
 - Hydro: 6.9%
 - Renewables: 5.7%

2021 data: Source BP Statistical Review of World Energy.
<https://www.energyinst.org/statistical-review>



Power ($W=J/s$): orders of magnitude



Denmark:

24 GW total power (ca 4.5 GW of electricity)

Power per captia



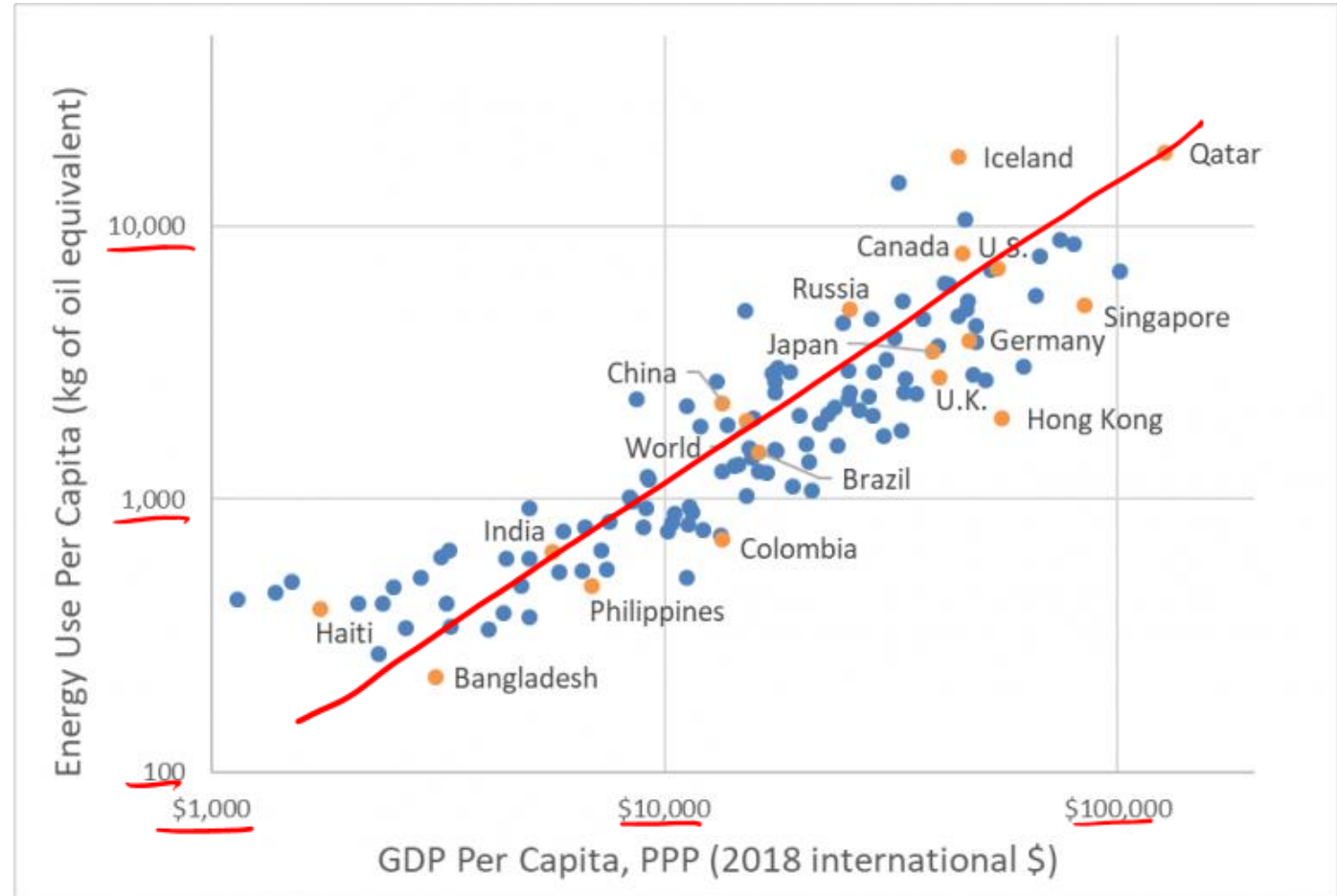
2.3 kW/person



4.1 kW/person

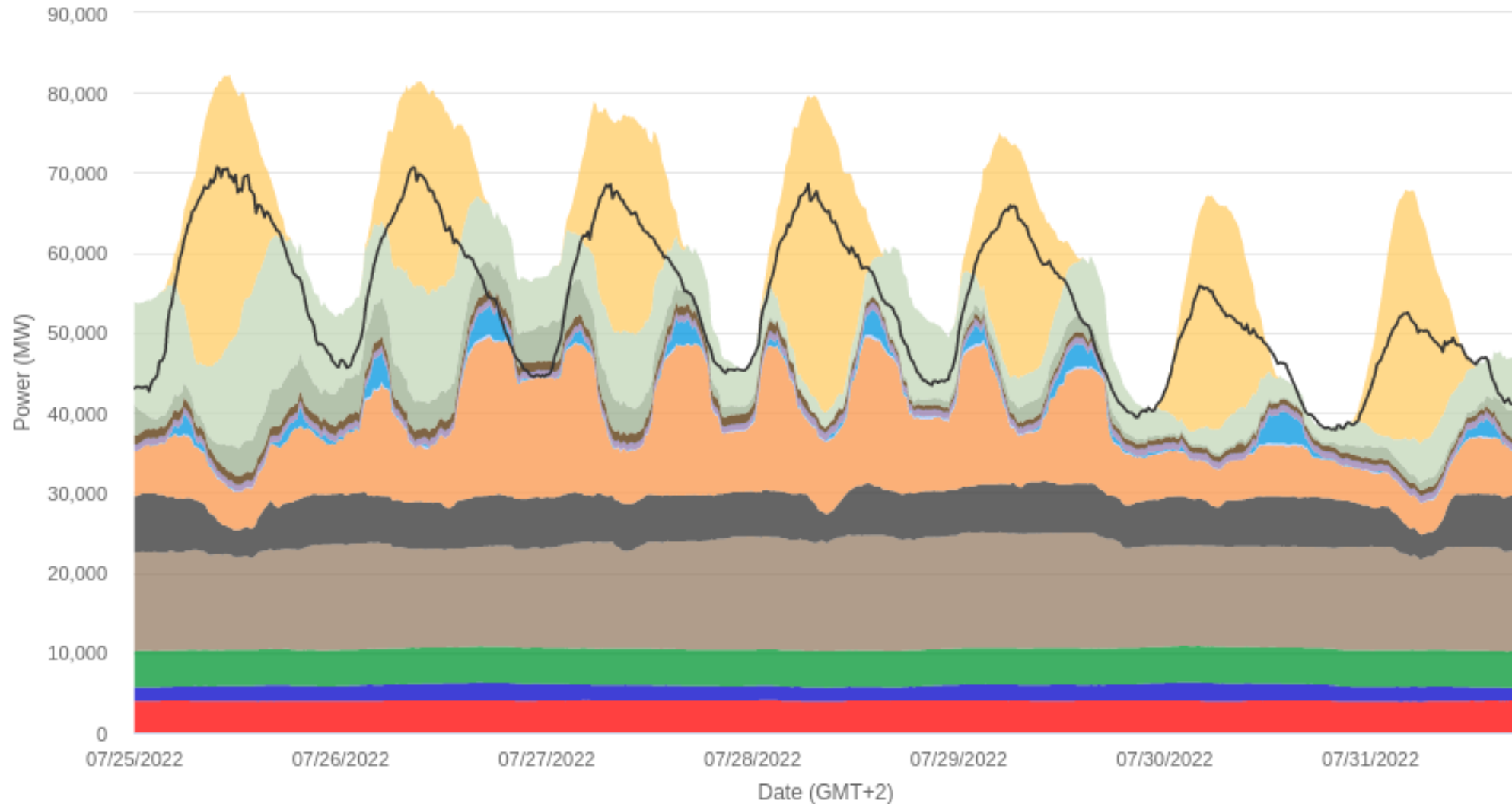


8.4 kW/person



Total net electricity generation in Germany in week 30 2022

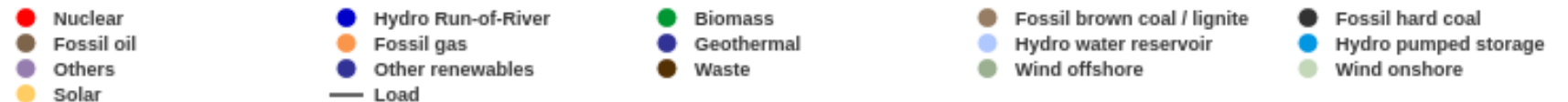
Energetically corrected values



**Renewable
energies need
storage**

Several time-scales
are relevant:

- Summer-winter
- Weeks
- Day-night
- Minute
- Sub-second



Motivation: Why store energy?

- Because the sun doesn't always shine and the wind doesn't always blow.
- Renewable energy sources *demand* energy storage in the system/grid.



Many technologies in play

Summer-winter

Solves:
Seasonal variations

Hydropower

(perhaps pumped hydro)

Mechanism

Physical, $E = mgh$

Main disadvantage

Requires mountains

Synthetic fuels

Also solves: Non-electrifiable necessities!

Mechanism

Chemical, via O_2

Main disadvantage

Very low energy conversion efficiency

Weeks

Solves:
Weather variations

Batteries

(perhaps redox-flow batteries)

Mechanism

Electrochemical

Main disadvantage

Expensive
Could depend on expensive/scarce elements

Compressed air (CAES)

Mechanism

Physical, $E = pV$

Main disadvantage

Requires natural underground caverns
Efficiency is so-so

Day-night (hours)

Solves:
Demand variations and solar on-off problem

Batteries

Mechanism

Electrochemical

Main disadvantage

Expensive
Could depend on expensive/scarce elements

Minute

Solves:
Grid frequency instability caused by industrial loads or clouds over solar

Flywheels

Mechanism

Physical, $E = 1/2 mv^2$

Main disadvantage

Expensive
Dangerous

Ultracapacitors

Mechanism

Electrochemical + electrostatic

Main disadvantage

Expensive

Sub-second

Solves:
**Grid shocks/spikes
Electrical faults & surges**

Flywheels

Mechanism

Physical, $E = 1/2 mv^2$

Main disadvantage

Expensive
Dangerous

Ultracapacitors

Mechanism

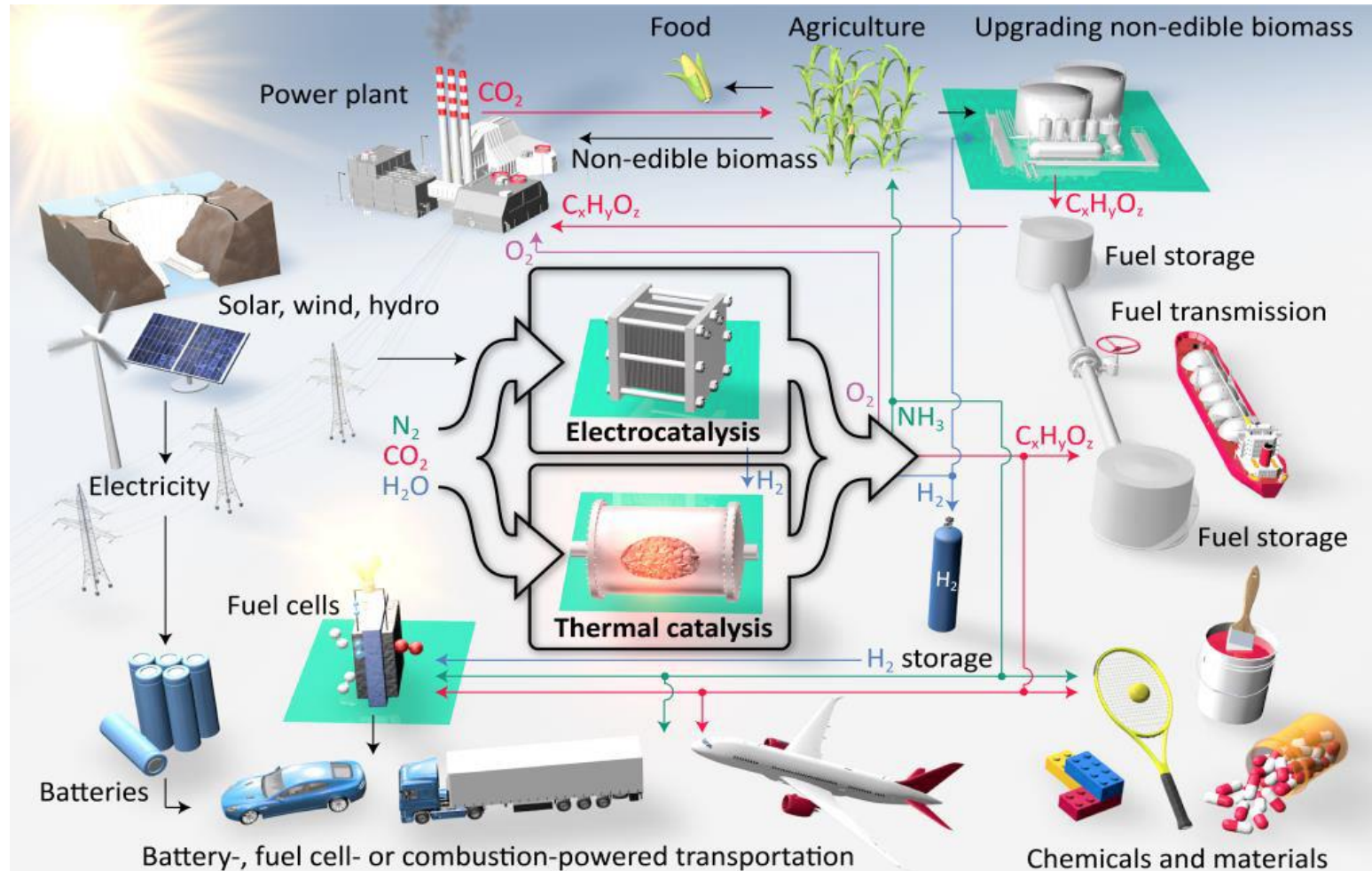
Electrochemical + electrostatic

Main disadvantage

Expensive

Big picture

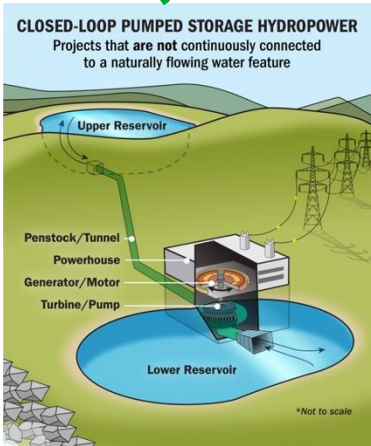
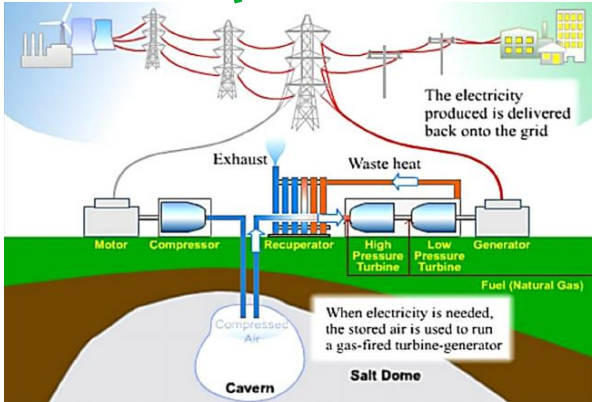
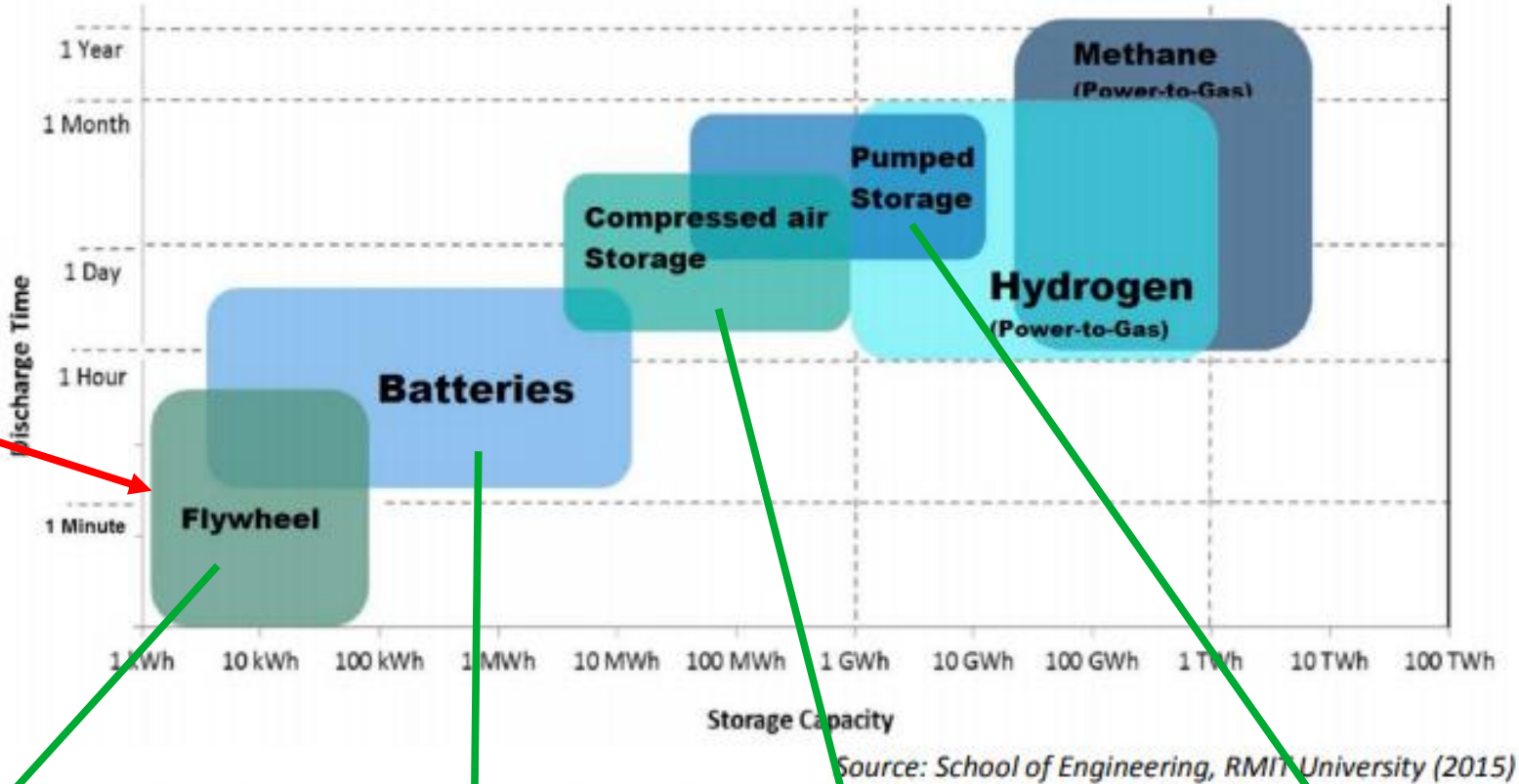
(ignoring the short term leveling)



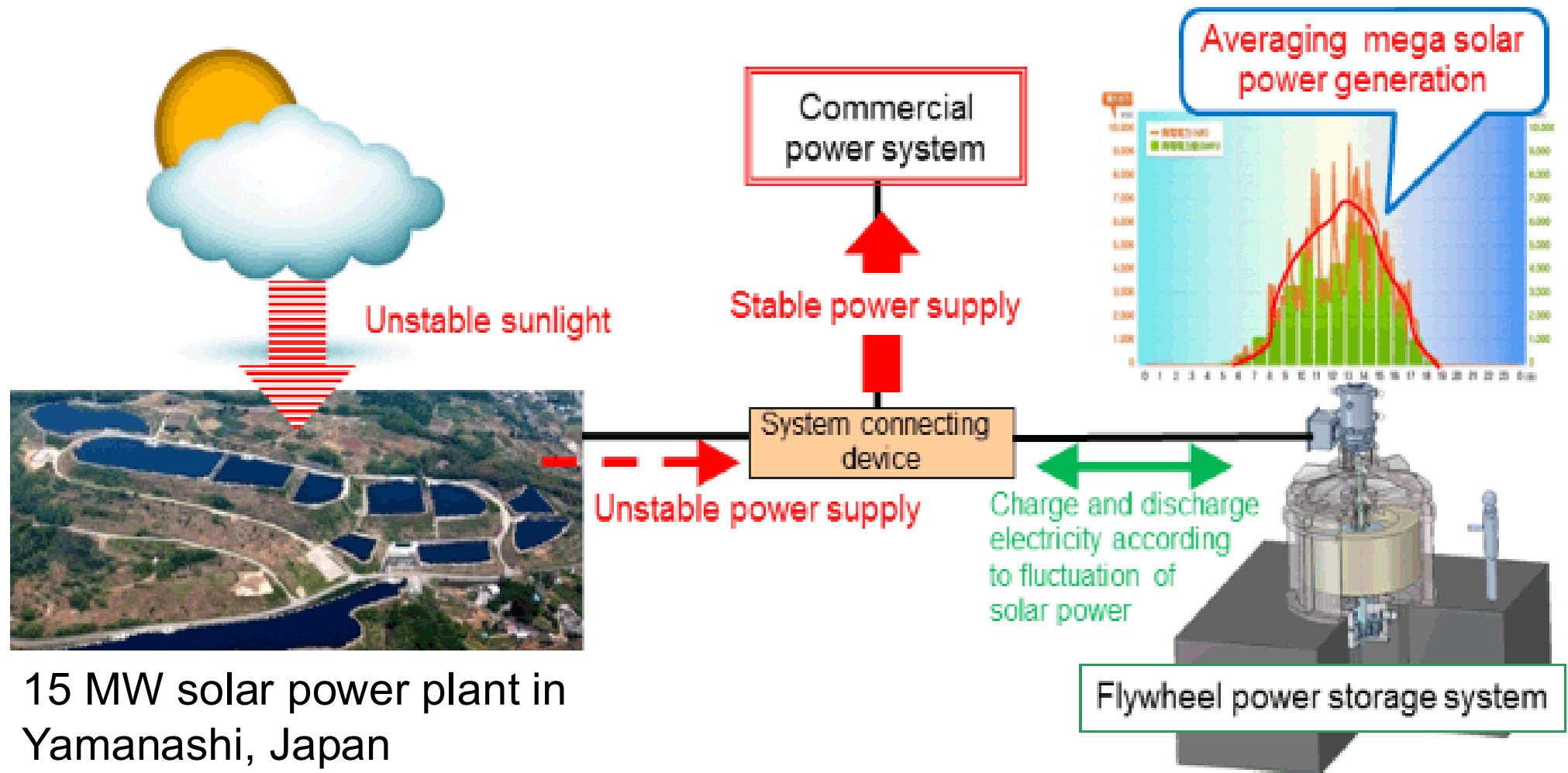
Flywheels are not relevant for large scale energy storage...

But they ARE relevant for short duration, high power energy storage and delivery.

Batteries + flywheels make sense with wind and solar



Example application - clouds over solar field



Basic principle: Q & A

Q: In what form does such a device store energy?

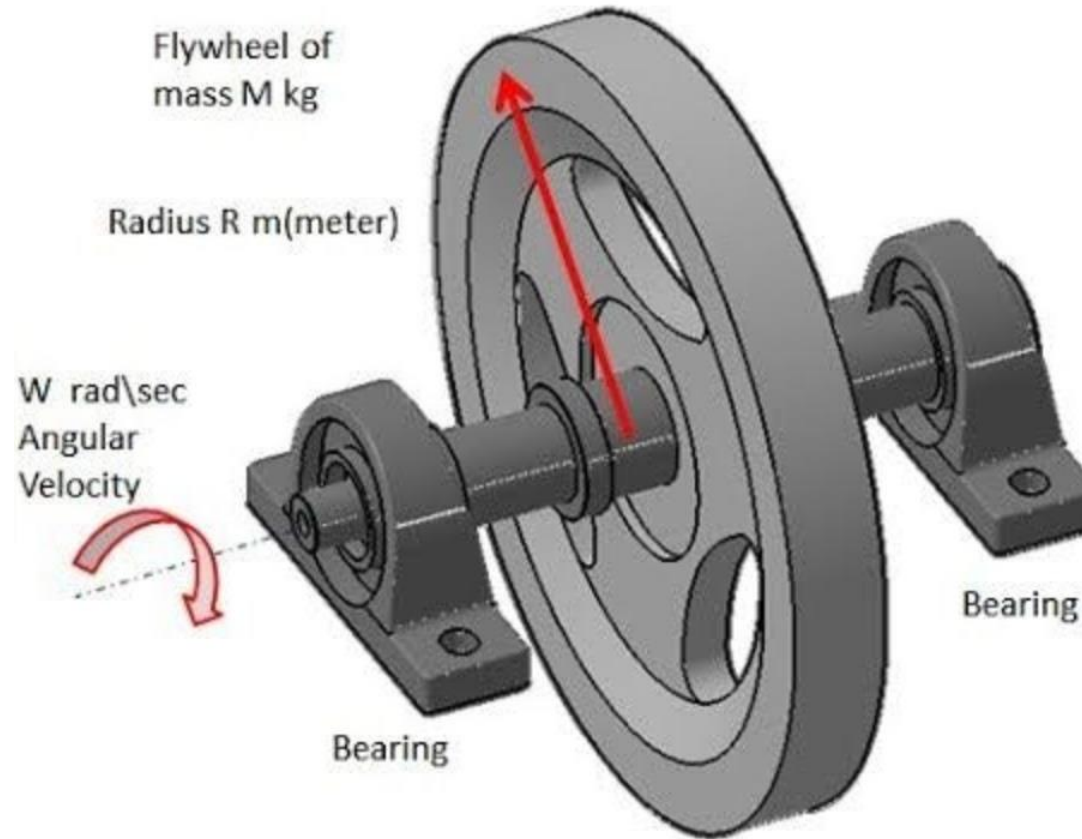
Kinetische Energi

$$E = \frac{1}{2} m v^2$$
$$= \frac{1}{2} I \omega^2$$

Q: What limits how long energy can be stored?

Friction

- × Kugellager
- × Luftwiderstand



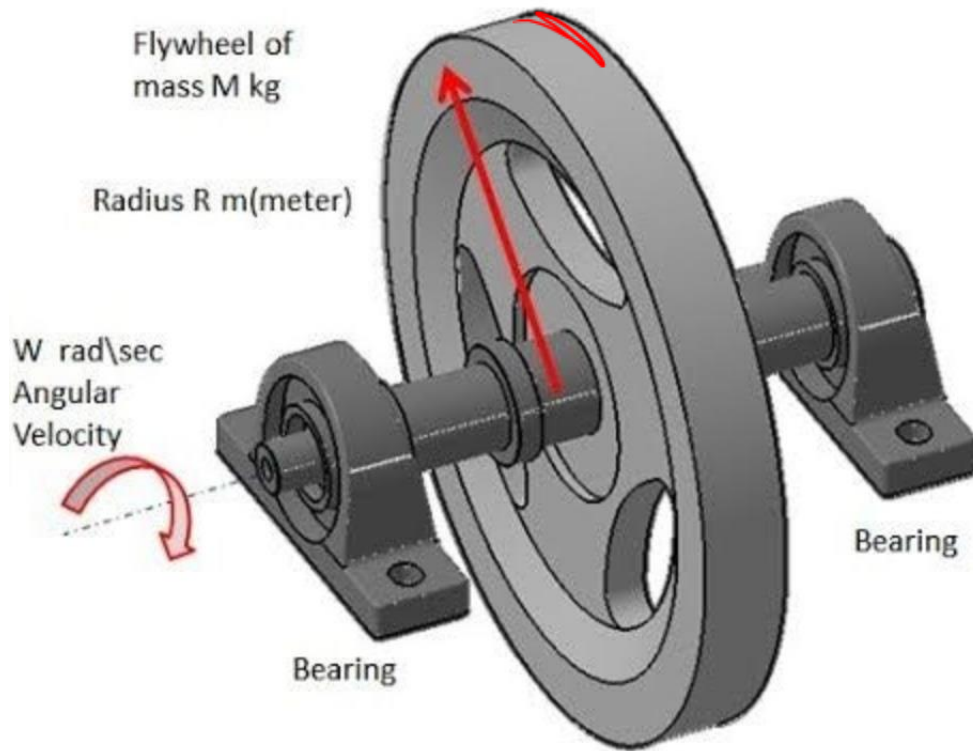
Q: What limits how much power can be delivered?

(stroke)
× generator.

Q: What limits how much energy can be stored?

- × Materialertrag
- × ω_{max} (max rotation)
- × v_{avg}
- × radius

Kinetic energy in the flywheel



$$E_{kin} = \frac{1}{2}mv^2$$

Minor problem:

v scales with the radius from the rotation axis:

$$v(r) = r\omega$$

Solution 1: Integration

$$E = \frac{1}{2} \int_0^R m(r)(v(r))^2 dr$$

Solution 2:

Look up the moment of inertia and use

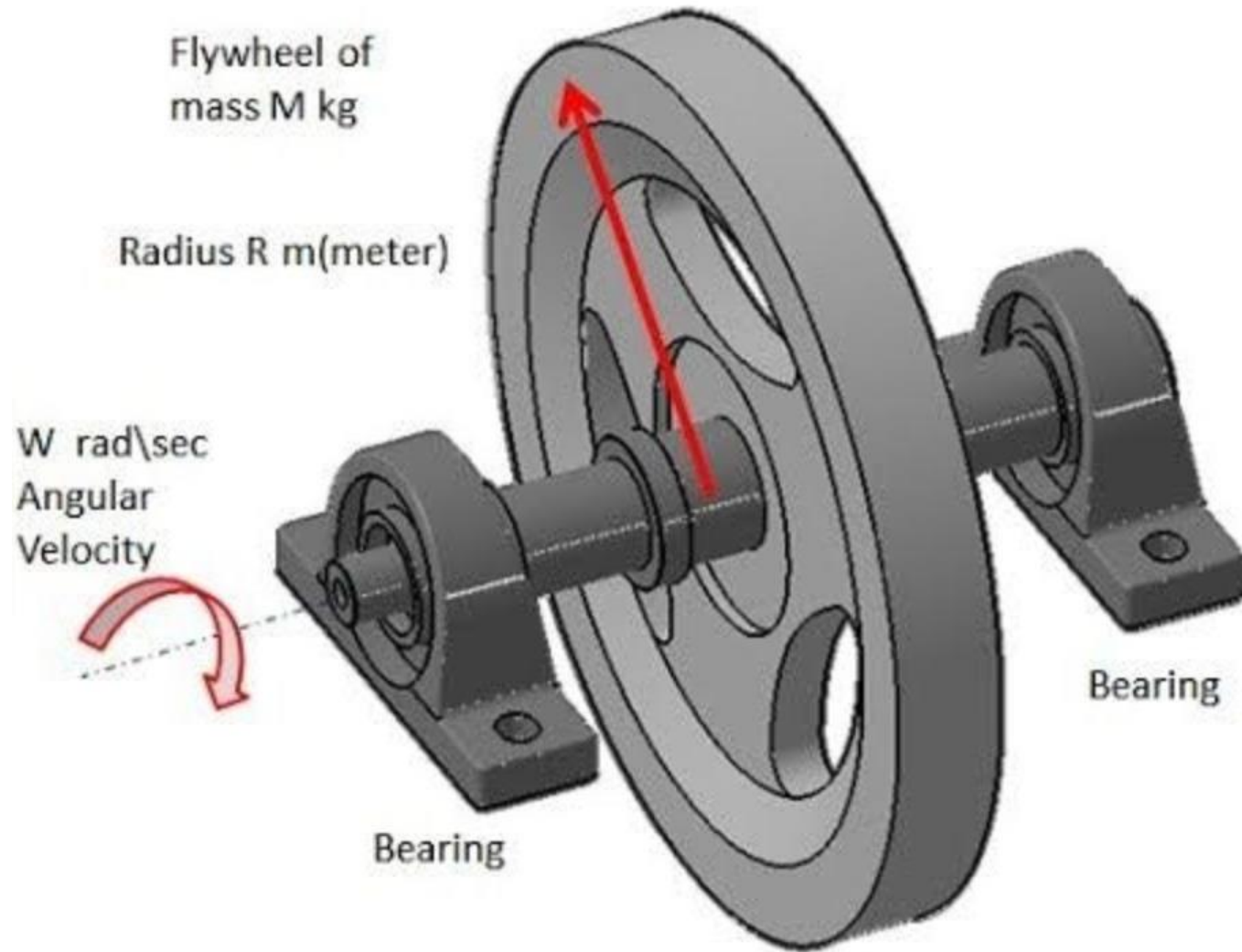
$$E_{kin} = \frac{1}{2}I\omega^2$$

Solution 3:

Consider a simple geometry where mass is at the same radius and therefore the same speed.

(This will be our approach)

Energy is limited by materials strength - but how, exactly?

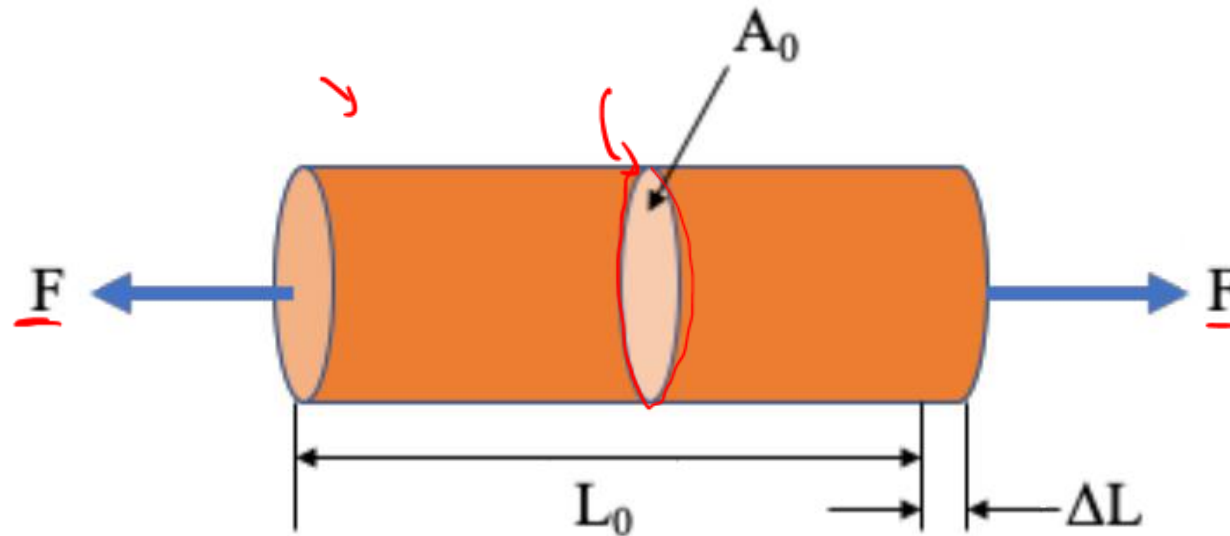


Can we calculate the maximum flywheel ω

- Strategy:
 - 1) Simplify the geometry as much as possible
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 - 4) Find how stress relates to rotational energy
 - 5) Find the limiting energy density of flywheels for various materials

First: What is material stress, σ ?

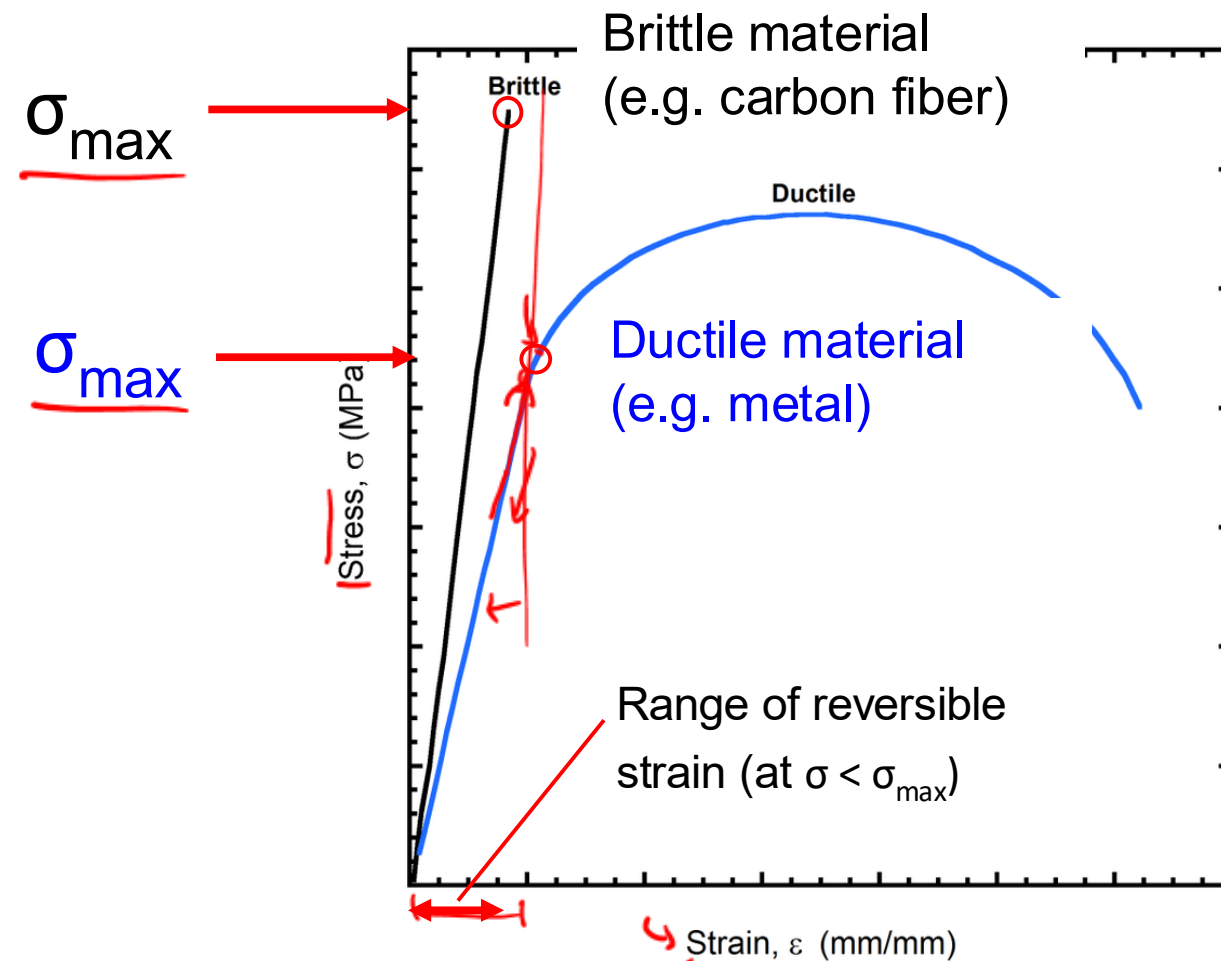
- Stress is force/area (unit: $\text{N/m}^2 = \text{Pa}$): $\sigma = F/A_0$
- Stress causes strain, $\underline{\epsilon}$ (material deformation): $\underline{\epsilon} = \underline{\Delta L}/\underline{L_0}$



What happens when stress is applied?

- Axial stress causes the material to stretch (strain)
- At $\sigma > \sigma_{\max}$, the material breaks (or deforms permanently)
- Our flywheel cannot be allowed to reach a stress above $\sigma_{\max}$

$$\sigma = F/A_0$$



Which materials have high $\sigma_{\max}$?



Titanium

Stahl

die mauert

Komposite (glas fiber, carbon...)

Some materials data

Typical tensile strengths of some materials

Material	Yield strength (MPa)	Ultimate strength (MPa)	Density (g/cm ³)
Structural steel ASTM A36 steel	250	400	7.8
Mild steel 1090	248	841	7.58
Human skin	15	20	2.2
2800 Maraging steel	2617	2693	8.00
AISI 4130 Steel, water quenched 855°C (1570°F), 480°C (900°F) temper	951	1110	7.85
Titanium 11 (Ti-6Al-2Sn-1.5Zr-1Mo-0.35Bi-0.1Si), Aged	940	1040	4.50
Steel, API 5L X65	448	531	7.8
Steel, high strength alloy ASTM A514	690	760	7.8
High-density polyethylene (HDPE)	26-33	37	0.95
Polypropylene	12-43	19.7-80	0.91
Stainless steel AISI 302 - Cold-rolled	520	860	8.19
Cast iron 4.5% C, ASTM A-48	130	200	
"Liquidmetal" alloy	1723	550-1600	6.1
Beryllium 99.9% Be	345	448	1.84
Aluminium alloy 2014-T6	414	483	2.8
Polyester resin (unreinforced)	55		
Polyester and Chopped Strand Mat Laminate 30% E-glass	100		
S-Glass Epoxy composite	2358		
Aluminium alloy 6061-T6	241	300	2.7
Copper 99.9% Cu	70	220	8.92
Cupronickel 10% Ni, 1.6% Fe, 1% Mn, balance Cu	130	350	8.94
Brass	200 +	550	
Tungsten		1510	19.25
Glass		33	2.53
E-Glass	N/A	1500 for laminates, 3450 for fibers alone	2.57
S-Glass	N/A	4710	2.48
Basalt fiber	N/A	4840	2.7
Marble	N/A	15	
Concrete	N/A	3	2.7

Typical tensile strengths of some materials

Material	Yield strength (MPa)	Ultimate strength (MPa)	Density (g/cm ³)
Carbon fiber	N/A	1600 for Laminate, 4137 for fiber alone	1.75
Carbon fiber (Toray T1000G)		6370 fibre alone	1.80
Human hair		380	
Bamboo		350-500	0.4
Spider silk (See note below)		1000	1.3
Darwin's bark spider silk	1652		
Silkworm silk	500		1.3
Polybenzoxazole (Zylon)		2700	1.56
Pine wood (parallel to grain)		40	
Bone (limb)	104-121	130	1.6
Nylon, type 6/6	45	75	1.15
Epoxy adhesive	-	12 - 30	-
Rubber	-	15	
Boron	N/A	3100	2.46
Silicon, monocrystalline (m-Si)	N/A	7000	2.33
Silicon carbide (SiC)	N/A	3440	
Ultra-pure silica glass fiber-optic strands		4100	
Sapphire (Al ₂ O ₃)	400 at 25°C 275 at 500°C 345 at 1000°C	1900	3.9-4.1
Boron Nitride Nanotube	N/A	33000	?
Diamond	1600	2800	3.5
Graphene	N/A	130000	1.0
First carbon nanotube ropes	?	3600	1.3
Colossal carbon tube	N/A	7000	0.116
Carbon nanotube (see note below)			
Carbon nanotube composites			
Iron (pure mono-crystal)			

<https://www.engineersedge.com/>
<https://matweb.com/>

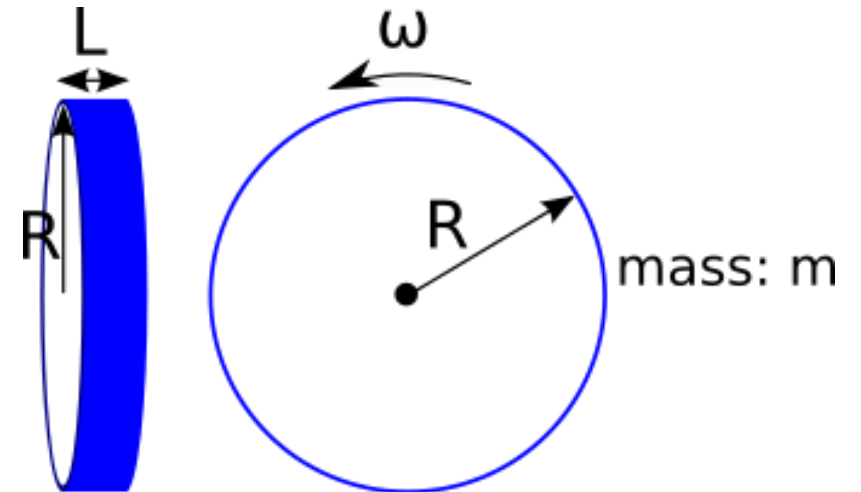
Can we calculate the maximum flywheel ω

- Strategy:
 - 1) **Simplify the geometry as much as possible**
 - 2) Assume constant circular motion
 - 3) Use knowledge of circ. mot. to find the material stress, σ
 - 4) Find how stress relates to rotational energy
 - 5) Find the limiting energy density of flywheels for various materials

Geometry

- Assume a simple geometry:
- The flywheel is just a thin hoop of material.
- Inner radius: R ,
- Outer radius: $R + \Delta R$
- Length (depth): L
- All mass moves at the same speed

$$R \gg \Delta R \Rightarrow \frac{(R + \Delta R)}{R} \sim 1$$



Can we calculate the maximum flywheel ω

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Circular motion

- Assume constant angular speed ω
- Gives rise to a centripetal acceleration a_c

$$\underline{a_c} = \underline{R} \underline{\omega^2}$$

- Consider a segment of the flywheel

$$\underline{dm} = \frac{m}{2\pi} \underline{d\theta}$$

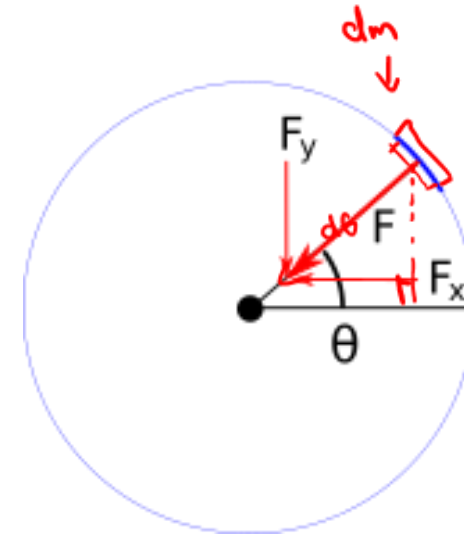
- Apply Newton's 2nd law to find F

$$\underline{|\vec{F}|} = \underline{a_c} \underline{dm}$$

- Find y- (and x) component of F

$$\underline{d\vec{F}_x} = \underline{a_c} \underline{\cos(\theta)} \underline{dm}$$

$$\underline{d\vec{F}_y} = \underline{a_c} \underline{\sin(\theta)} \underline{dm}$$



small segment:
with angle $d\theta$
mass: dm

Can we calculate the maximum flywheel ω

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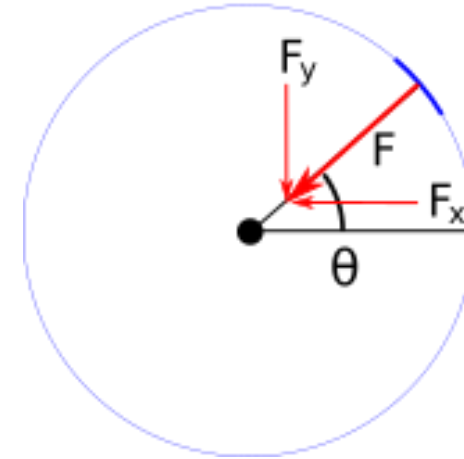
Calculate the force due to circ. mot.

$$a_c = R\omega^2 \quad dm = \frac{m}{2\pi} d\theta$$

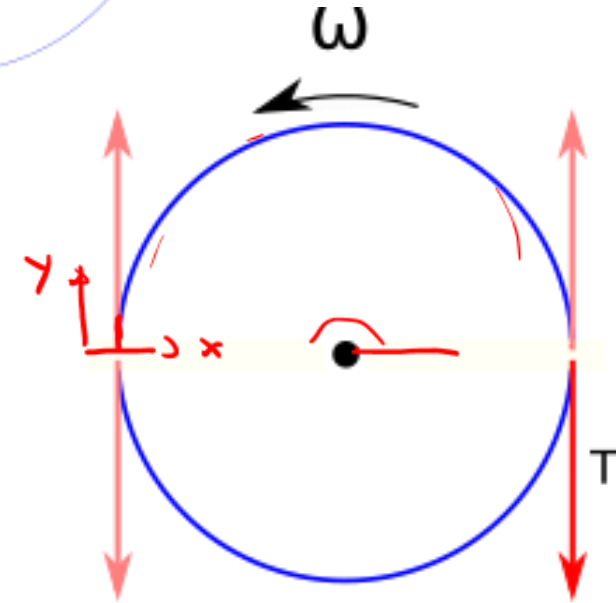
$$|\vec{F}| = a_c dm$$

$$\rightarrow d\vec{F}_y = a_c \sin(\theta) dm$$

$$dF_y = a_c \sin(\theta) \frac{m}{2\pi} d\theta$$



small segment:
with angle $d\theta$
mass: dm



Calculate the force due to circ. mot.

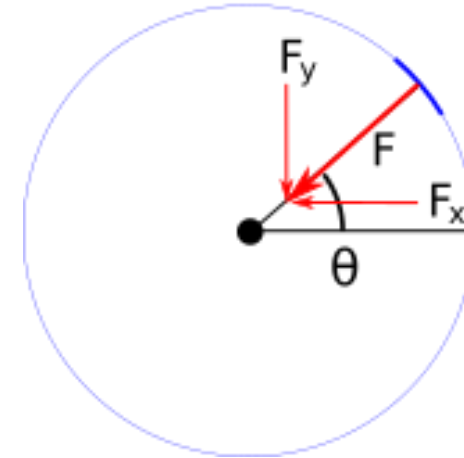
$$\underline{a_c} = \underline{R\omega^2} \quad \overrightarrow{dF_y} = a_c \frac{m}{2\pi} \sin(\theta) d\theta \quad \leftarrow$$

$$\begin{aligned} 2T &= \int_{\text{top wheel}} dF_y = \int_0^{\pi} a_c \frac{m}{2\pi} \sin \theta d\theta \\ &= a_c \frac{m}{2\pi} \int_0^{\pi} \sin \theta d\theta = a_c \frac{m}{2\pi} [-\cos \theta]_0^{\pi} \\ &= a_c \frac{m}{2\pi} (-\cos(\pi) - (-\cos(0))) \\ &= a_c \frac{m}{2\pi} (-(-1) - (-1)) \end{aligned}$$

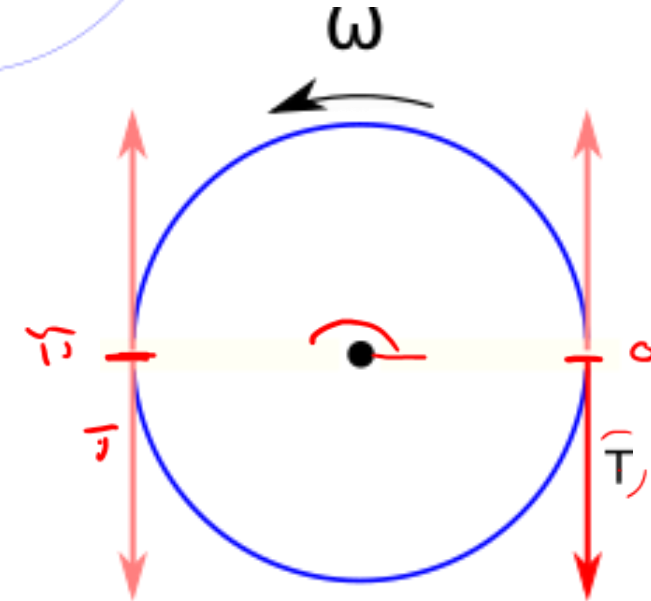
$$2T = a_c \frac{m}{2\pi} \cdot 2$$

$$\boxed{T = \frac{Rm}{2\pi} \cdot \omega^2}$$

↑ ↑
Kraft



small segment:
with angle $d\theta$
mass: dm



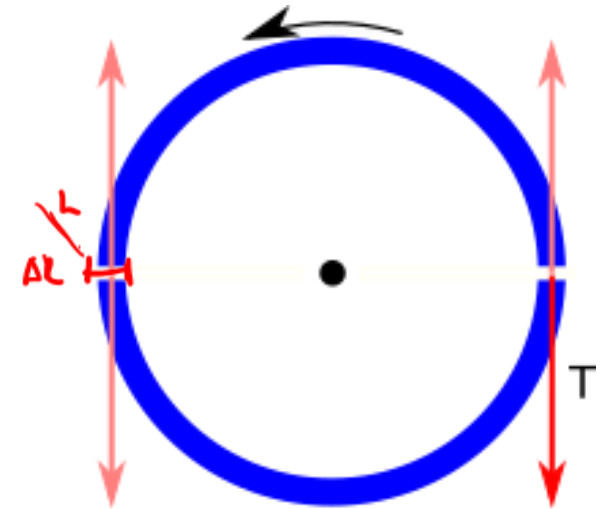
Calculate stress in the wheel

$$T = \frac{mR\omega^2}{2\pi}$$

$$\sigma = \frac{F}{A}$$

$$A = (\Delta R)L$$

$$\sigma = \frac{T}{A} = \frac{mR\omega^2}{2\pi A}$$



Length in the (z-direction) is L

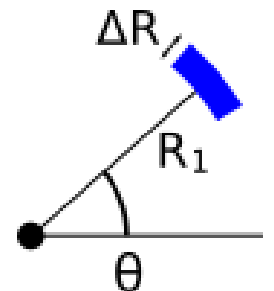
Volume of the flywheel is $V \approx 2\pi R A \Rightarrow \underline{m} = \underline{\rho} V = \underline{\rho} 2\pi R A$

Density



$$\sigma = \frac{\rho 2\pi R A \cdot R \omega^2}{2\pi A} = \rho \cdot R^2 \cdot \omega^2$$

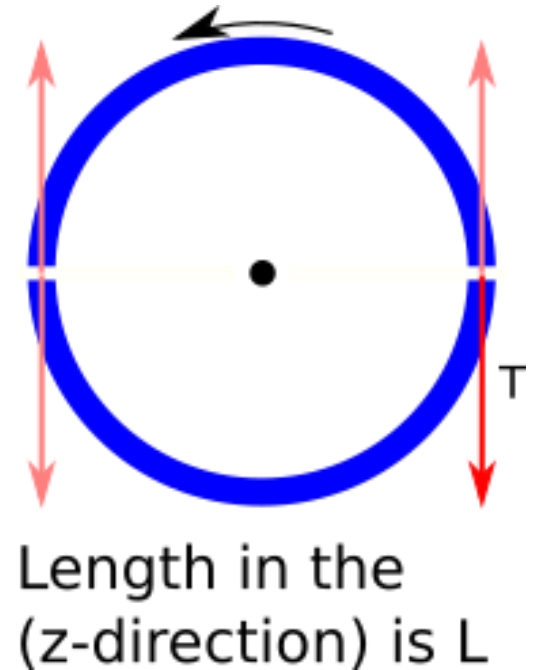
$$\sigma = \rho R^2 \omega^2$$



Stress in the flywheel

$$\sigma = \rho R^2 \omega^2$$

- The material stress, σ , scales with
 - the material density
 - the square of the flywheel radius
 - the square of the rotational speed



Can we calculate the maximum flywheel ω

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How energy relates to stress

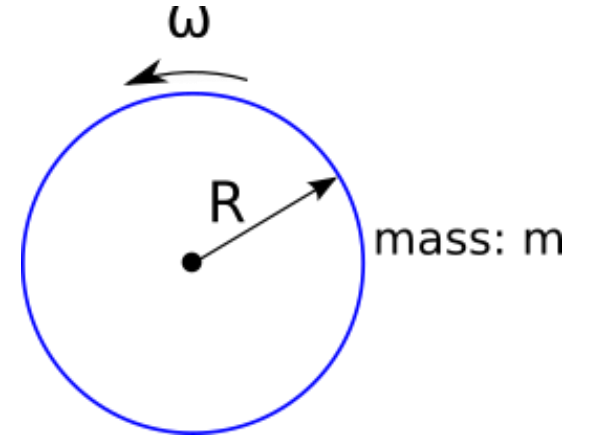
$$\boxed{\underline{\sigma} = \underline{\rho} R^2 \omega^2} \rightarrow \frac{\underline{\sigma}}{\underline{\rho}} = \underline{R^2 \cdot \omega^2}$$

$$\begin{aligned} v &= R\omega \\ \underline{E_k} &= \frac{1}{2} m v^2 = \frac{1}{2} m \underline{R^2 \omega^2} \end{aligned}$$

$$E_{kin} = \frac{1}{2} m \frac{\underline{\sigma}}{\underline{\rho}} \quad \text{c/B}$$

$$E_{kin} = \frac{1}{2} \underline{\sigma} \cdot V$$

$$\Rightarrow \boxed{\frac{E_{kin}}{V} = \frac{\underline{\sigma}}{2}}$$



Can we calculate the maximum flywheel ω

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Limiting energy in a flywheel

$$\frac{E_{kin}}{V} = \frac{\sigma}{2}$$

Remarkable result: The energy per volume depends only on the stress in the material!

I.e. it does not depend on material density, flywheel radius or rotational speed!

Conclusion: The maximum volumetric energy density of a flywheel depends ONLY on the maximum allowable stress.

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Material	Yield strength (MPa)	Ultimate strength (MPa)	Density (g/cm ³)
Structural steel ASTM A36 steel	250	400	7.8
Mild steel 1090	248	841	7.58
Human skin	15	20	2.2
2800 Maraging steel	2617	2693	8.00
AISI 4130 Steel, water quenched 855°C (1570°F), 480°C (900°F) temper	951	1110	7.85
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Cupronickel 10% Ni, 1.6% Fe, 1% Mn, balance Cu	130	350	8.94
Brass	200 +	550	
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Glass	33		2.53
E-Glass	N/A	1500 for laminates, 3450 for fibers alone	2.57
S-Glass	N/A	4710	2.48
Basalt fiber	N/A	4840	2.7
Marble	N/A	15	
Concrete	N/A	3	2.7

Examples from the previous table:

Structural Steel, $\sigma_{max} = 250 \text{ MPa} \Rightarrow E/V \leq 125 \text{ MJ/m}^3$

Carbon fiber, $\sigma_{max} = 1600 \text{ MPa} \Rightarrow E/V \leq 800 \text{ MJ/m}^3$

Break

12 min

11 min

10 min

9 min

8 min

7 min

6 min

5 min

4 min

3 min

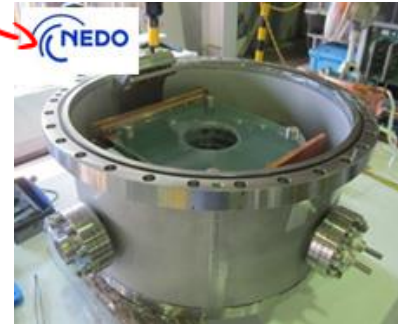
2 min

1 min

Time!

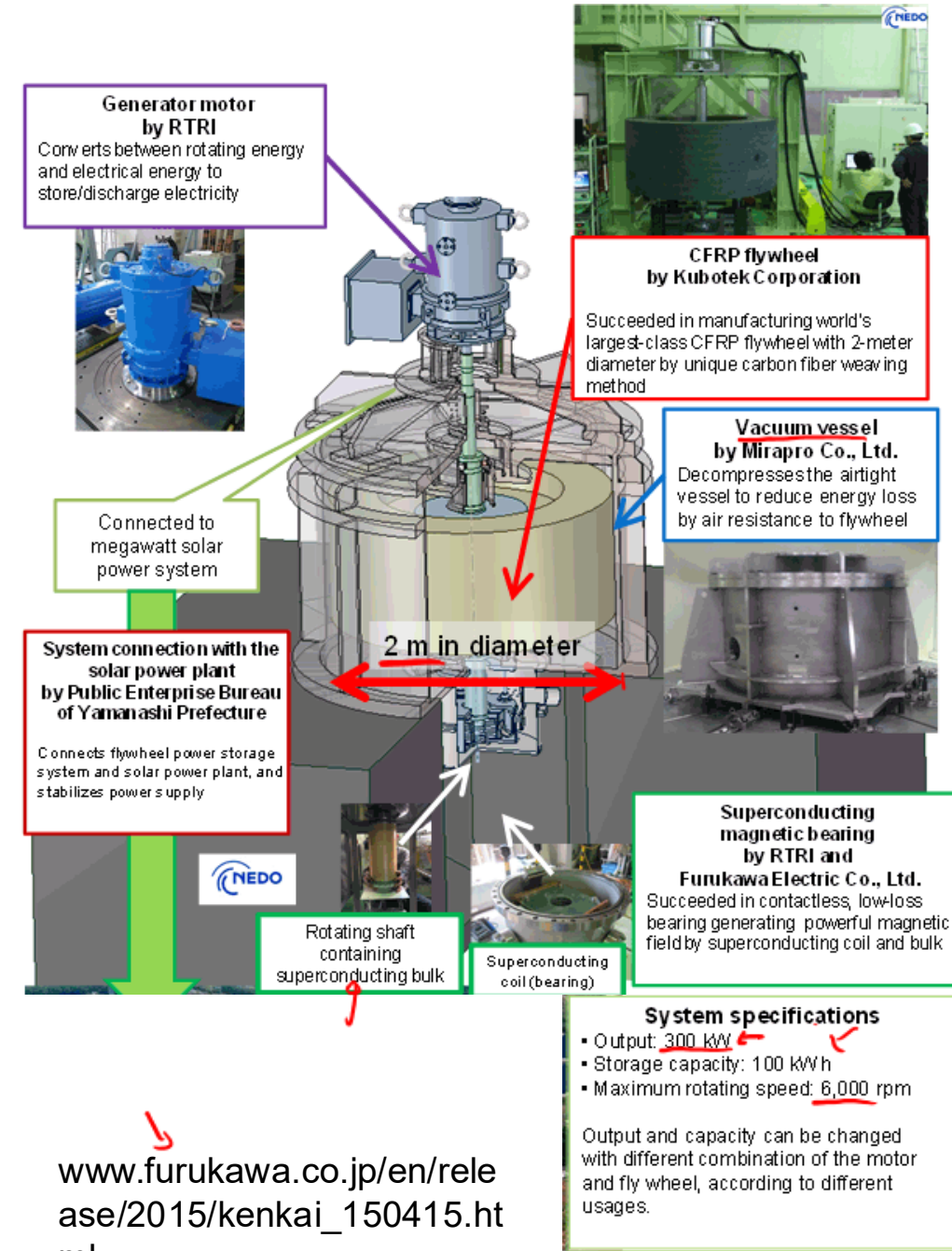
Real world flywheels?

- Two sources of friction:
 - Friction in the bearings
 - Air drag



Superconducting bearing

2 meter diameter carbon fiber flywheel



Power Banks versus flywheel

If a power bank is marked as 10,000mAh 3.8V (38Wh), that means its battery capacity is 10,000mAh, the voltage of the battery core is 3.8V, and the battery core energy is 38Wh

$$\text{Energy: } \underline{38 \text{ Wh}} = \underline{38} * \underline{60} * \underline{60} \text{ J} = \underline{0.137 \text{ MJ}}$$

$$\rightarrow \frac{\text{Energy}}{\text{mass}} = \frac{0.137 \text{ MJ}}{220 \text{ g}} = \underline{620 \text{ J/g}}$$

Teknisk info:

- Batteriklasse: Lithium Polymer 3,8V 10000mAh 38Wh
- Driftstemperatur: 5° C til 35° C
- Dimensioner (LxHxB): 25 x 150 x 90 mm
- Vægt: 220 g



Power Banks versus flywheel

Power Bank : $\frac{\text{Energy}}{\text{mass}} = \frac{0.137 \text{ MJ}}{220 \text{ g}} = \underline{620 \text{ J/g}}$

Flywheel :

Structural Steel $\frac{E}{V} \leq \underline{125 \frac{\text{J}}{\text{cm}^3}}$, $\text{density} = \underline{7.8 \frac{\text{g}}{\text{cm}^3}} \Rightarrow \frac{\text{Energy}}{\text{mass}} = \underline{16 \text{ J/g}}$

Carbon fiber $\frac{E}{V} \leq \underline{800 \frac{\text{J}}{\text{cm}^3}}$, $\text{density} = \underline{1.75 \frac{\text{g}}{\text{cm}^3}} \Rightarrow \frac{\text{Energy}}{\text{mass}} = \underline{460 \text{ J/g}}$

Quiz

Power Banks/flywheel versus mars bar

3 min
2 min
1 min

Time!

Power Bank : $\frac{\text{Energy}}{\text{mass}} = \frac{0.137 \text{ MJ}}{220 \text{ g}} = \underline{620 \text{ J/g}}$

Flywheel :

Structural Steel $\frac{E}{V} \leq 125 \frac{\text{J}}{\text{cm}^3}$, $\text{density} = 7.8 \frac{\text{g}}{\text{cm}^3} \Rightarrow \frac{\text{Energy}}{\text{mass}} = 16 \text{ J/g}$

Carbon fiber $\frac{E}{V} \leq 800 \frac{\text{J}}{\text{cm}^3}$, $\text{density} = 1.75 \frac{\text{g}}{\text{cm}^3} \Rightarrow \frac{\text{Energy}}{\text{mass}} = 460 \text{ J/g}$

What is the **energy / mass** in a mars bar?

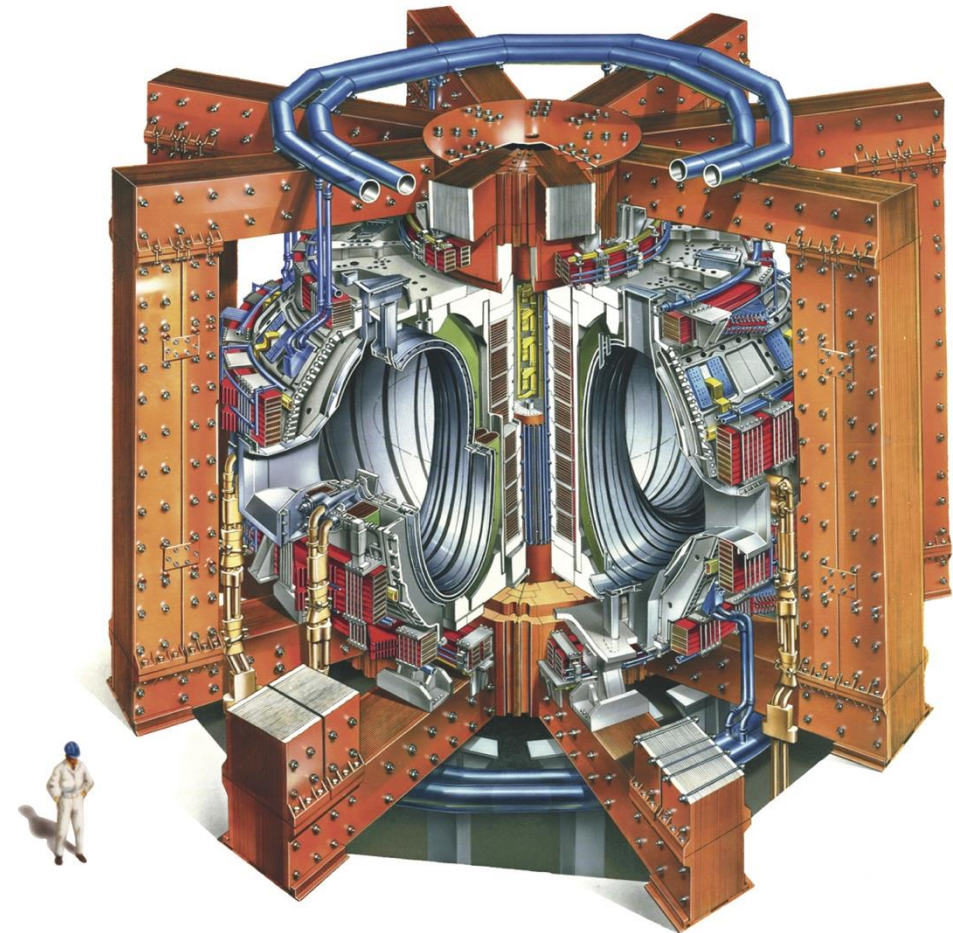
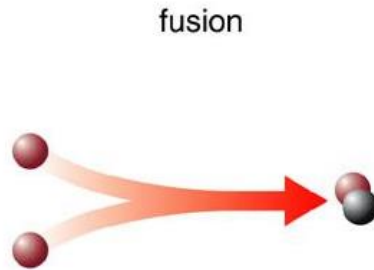
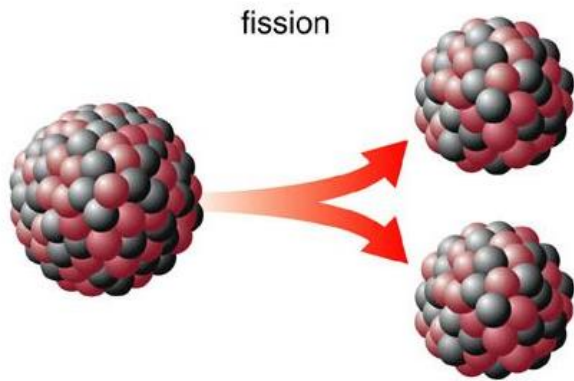


Kemisk Energi

$$\frac{E}{\text{mass}} = \frac{960 \text{ kJ}}{51 \text{ g}} = 19 \text{ kJ/g} = 19,000 \text{ J/g}$$

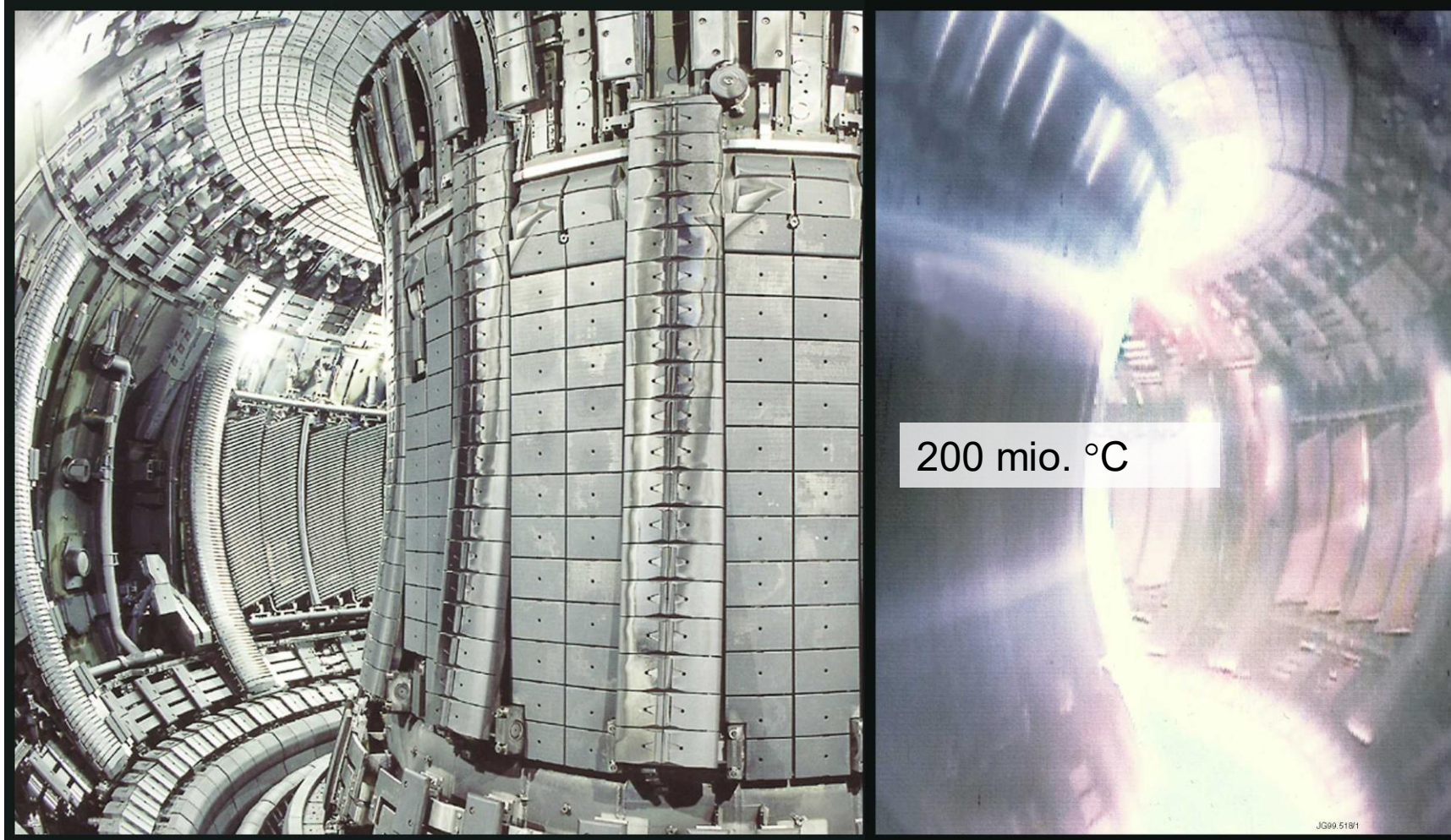
$$\text{FTP} : 280 \text{ W} \cdot 60 \cdot 60 = \underline{\underline{1008 \text{ kJ}}}$$

Verdens største tokamak: JET (Joint European Torus)



1983
ved Oxford, UK

Fusionsplasma i JET



Avedøreværket

Den **samlede produktionskapacitet** for Avedøreværkets to blokke er på **793 MW el** og **918 MJ/s varme**. Avedøreværket forsyner 200.000 husstande med varme og producerer ca. **30 % af elforbruget på Sjælland**.



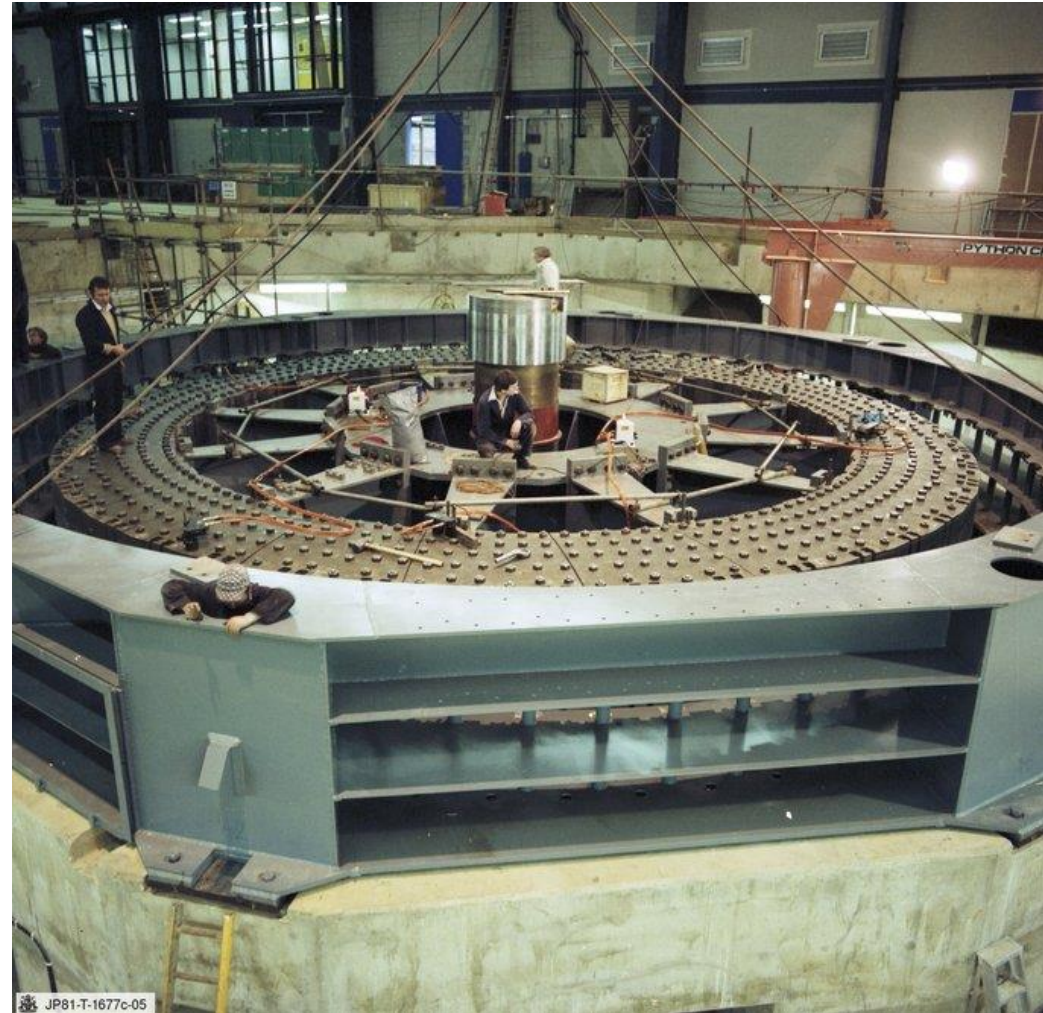
Fusion reactor Joint European Torus (JET)

JET has **two 775 t flywheels** (installed in 1981) that spin up to 225 rpm.

Each flywheel **stores 3.75 GJ** and can deliver at up to **400 MW**

2 x 400 MW ~ Avedøreværkets **793 MW el**

Duration of a single experimental is 60 second



Kursusoverblik

Overblik over forårets forelæsninger, **foreløbig** plan:

Uge 1: Kinematik 1D (Kap. 3)
Uge 2: Kinematik 2D (Kap. 4)
Uge 3: Usikkerhed
Uge 4: Eksperimenter
Uge 5: Kræfter 1 (Kap. 5)
Uge 6: Kræfter 2 (Kap. 6)
Uge 7: Bevægelsesligninger (Kap. 15)
Uge 8: Arbejde Og Energi (Kap. 7)
Påske (30/3)
Påske (6/4)
Uge 9: Energibevarelse (Kap. 8)
Uge 10: Stød Og Impulsbevarelse (Kap. 9)
Uge 11: Rotation 1 (Kap. 10)
Uge 12: Rotation 2 (Kap. 11)
Uge 13: Energitema 1

Lærebogen er University Physics fra OpenStax, der er gratis tilgængelig på <https://openstax.org/details/books/university-physics-volume-1> og <https://openstax.org/details/books/university-physics-volume-2>.

Link til boghandel:

https://polyteknisk.dk/home/dtu/soege_resultat?utf8=%E2%9C%93&q=10060&b=20&st=c&code=&exact_match=false

Gode studie vaner:

- Læs kapitel inden forelæsning,
- Forelæsning
- Gruppe regning (lær af dine medstuderende)
- Færdiggør gruppe øvelser hjemme.

Repetition (3 gange) & øvelse er nøglen!